

B.Tech Degree VI Semester Examination April 2012

ME 602 DYNAMICS OF MACHINERY

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Explain the effect of Pin Joint Friction in the analysis of slider crank mechanism.
 (b) Explain the concept of equivalent dynamical system.
 (c) Derive an expression for gyroscopic couple on a spinning disc.
 (d) What are the effects of friction and of adding a central weight to the sleeve of a watt governor?
 (e) Explain the 'direct and reverse crank' method for determining unbalanced forces in radial engines.
 (f) Explain the terms 'static balancing' and 'dynamic balancing'. State the necessary conditions to achieve them.
 (g) It is stated that the speed at which a belt or rope should be run to transmit maximum power is that at which the maximum allowable tension is three times the centrifugal tension in the belt or rope at that speed. Prove the statement.
 (h) What are the leading and trailing shoes of an internal expanding shoe brake?

PART B

(4 x 15 = 60)

- II. Construct free body diagrams for each elements of the linkage and determine the magnitude and direction of the forces acting on each part. The dimensions of various links are as follows : AB = 5 cm, BC = AD = 30 cm, CD = 17 cm and $\angle BAD = 60^\circ$. Link AD is fixed. The frictional forces may be neglected. A torque of magnitude 3000 N cm is acting on link CD in the anticlockwise direction. What input torque must be applied to link AB for equilibrium? (15)

OR

- III. (a) Explain the principle of super position. (3)
 (b) Explain the terms shaking forces and shaking moments. (3)
 (c) In a double acting vertical steam engine running at 360 rpm, cylinder diameter is 25 cm, stroke is 30 cm, diameter of piston rod is 3.75 cm and length of connecting rod is 60 cm. When the crank has moved 120° from TDC, the pressure of the steam at cover end is $35 \times 10^4 \text{ N/m}^2$ and at the crank end is $3 \times 10^4 \text{ N/m}^2$. If the mass of the reciprocating parts is 45 Kg, find: (i) piston effort (ii) turning moment on the crank shaft for the given crank position. (9)

- IV. A multicylinder engine is to run at a speed of 600 rpm. On drawing the crank effort diagram to scale of 1 cm – 2500 Nm and 1 cm – 30° , the areas above and below the mean torque line in sq. cm are as follows : -1.6, -1.72, +1.68, -1.91, +1.97, -1.62. The speed is to be kept within 2% of the mean speed of the engine. Calculate the necessary moment of inertia of the flywheel. Determine the suitable dimensions of the rectangular flywheel rim if the breadth is twice its thickness. The density of CI is 7.25 gm/cm^3 and hoop stress is 600 N/cm^2 . (15)

OR

- V. (a) In a spring controlled governor, the curve of controlling force is a straight line. When balls are 400 mm apart, the controlling force is 1200 N and when 200 mm apart, the controlling force is 450 N. At what speed will the governor run when the balls are 250 mm apart? What initial tension on the spring would be required for isochronism and what would then be the speed? The mass of each ball is 9 Kg. (7)

(P.T.O.)

- (b) The turbine rotor of a ship weighing 35 kN has a radius of gyration of 45 cm. If rotates at a uniform speed of 3000 rpm clockwise when viewed from the stern, determine the gyroscopic couple and its effects upon the ship under the following conditions :

- (i) when the ship is taking a curve of radius 100 m to the left at a cruising speed of 35 Km/hr.
- (ii) when the ship is pitching in SHM the bow falling with its maximum velocity. The period of pitching is 40 seconds and the total angular displacement between the two extreme positions of pitching is 12° .

(8)

- VI. (a) Derive expressions for the maximum variation in tractive effort and also for swaying couple in an uncoupled locomotive.

(5)

- (b) A 4 cylinder in line marine oil engine has cranks at angular displacement of 90° . The outer cranks are 3.5 m apart and inner cranks are 1.5 m apart. The inner cranks are placed symmetrically between the outer cranks. The length of each crank is 500 mm. If the engine runs at 120 rpm and the mass of reciprocating parts of each cylinder is 8 kN, find the firing order of the cylinders for the best primary balancing forces of reciprocating masses. Determine the maximum unbalanced 1° couple for the best arrangement.

(10)

OR

- VII. (a) Explain the methods for balancing of V engines.

(5)

- (b) A 90° twin cylinder V engine has a stroke of 12.5 cm. The connecting rods are equal in length and measure 24.5 cm. The crank pin and crank webs are equivalent to 10 N at crank radius and each piston weighs 8 N. The weight of each connecting rod is 15 N and the CG of each connecting rod is 4.5 cm from the crank pin centre.

Show that the effect of the revolving mass and the 1° effect of the reciprocating masses may be balanced by a revolving weight. Find its magnitude and position if the distance of the CG of the balance weight from the crank shaft centre line is 4.5 cm. What is the nature and magnitude of the resultant secondary forces, when the crank makes 1,600 rpm?

(10)

- VIII. (a) A vehicle is moving on a level road at a speed of 36 Km/h. Its CG lies at a distance of 0.6 m from the ground level. The wheel base is 2.4 m and the distance of CG from the rear wheels is 0.9 m. Find the distance travelled by the vehicle before coming to rest when brakes are applied, (i) to the rear wheels (ii) to the front wheels (iii) to all the four wheels. The coefficient of friction between the tyres and the road surface is 0.45.

(8)

- (b) A cone clutch with cone angle 20° is to transmit 7.5 KW at 750 rpm. The normal intensity of pressure between the contact faces is not to exceed 0.12 N/mm^2 . The coefficient of friction is 0.2. If the face width is $1/5^{\text{th}}$ of mean diameter, find : (i) the main dimensions of the clutch (ii) axial force required while running.

(7)

OR

- IX. (a) A rope drive transmits 75 KW through a 1.5 m diameter, 45° grooved pulley rotating at 200 rpm. The coefficient of friction between the ropes and the pulley grooves is 0.3 and the angle of lap is 160° . Each rope has a mass of 0.6 Kg/m and can safely take a pull of 800 N. Taking centrifugal tension into account determine : (i) the number of ropes required for the drive (ii) initial rope tension.

(8)

- (b) Two shafts whose centres are 1 m apart are connected by V – belt drive. The driving pulley is supplied with 100 KW and has an effective diameter of 300 mm. It runs at 1000 rpm while the driven pulley runs at 375 rpm. The angle of the groove on the pulley is 40° . The permissible tension is 400 mm^2 . Cross sectional area belt is 2.1 MPa. The density of belt is 1100 Kg/m^3 . The coefficient of friction between belt and pulley is 0.28. Estimate the number of belts required.

(7)